

# Vishesh Saluja

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## Education

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|-----------------------------------------------------------------------------------------------------|---------------------------|
| <b>George Mason University</b> , Master of Science in Computer Science<br><i>Fairfax, VA, USA</i>   | May 2025                  |
| <b>BVCOE – GGSIPU</b> , Bachelor of Technology in Information Technology<br><i>New Delhi, India</i> | GPA: 8.7/10.0    Jul 2023 |

## Experience

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| <b>Software Developer &amp; Research Assistant</b> , <i>Geoweaver – NASA/ESIP</i> <ul style="list-style-type: none"><li>Led 4 interns on ML workflows for geospatial data.</li><li>Added support for PostgreSQL, MySQL, and H2 with CLI integration.</li><li>Implemented an in-app code editor and improved UI responsiveness by 30%.</li><li>Reduced processing time by 25% in Snow/Ozone models.</li><li>Ran large-scale workflows on Hopper supercomputer with 20% speed-up.</li></ul> | Feb – Oct 2024 |
| <b>Data Science Intern</b> , <i>QBurst Technologies</i> <ul style="list-style-type: none"><li>Improved BigQuery query performance by 15%.</li><li>Conducted training and documentation improving team output by 10%.</li></ul>                                                                                                                                                                                                                                                            | Jul – Aug 2022 |
| <b>ML Intern</b> , <i>DynPro</i> <ul style="list-style-type: none"><li>Built models improving accuracy by 18%.</li><li>Delivered regression and clustering results boosting success rate by 20%.</li></ul>                                                                                                                                                                                                                                                                                | Apr – Jun 2022 |
| <b>Software Engineer Intern</b> , <i>Accops Systems</i> <ul style="list-style-type: none"><li>Worked on backend logic using Python and C++.</li><li>Reduced bug resolution time by 30%.</li></ul>                                                                                                                                                                                                                                                                                         | Jan – Mar 2022 |
| <b>Cloud Trainee</b> , <i>Tech Explica</i> <ul style="list-style-type: none"><li>Deployed multiple websites on AWS.</li><li>Increased deployment efficiency by 20%.</li></ul>                                                                                                                                                                                                                                                                                                             | Jul – Aug 2021 |

## Publication

- Socioeconomic and Industrial Factors Influencing U.S. Ozone Pollution – NASA ESIP (AGU).
- Snow Water Equivalent Estimation Using ML & Remote Sensing – NASA ESIP.

## Projects

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| <b>Mason Critique</b> – <i>Awarded 2nd Place</i> <ul style="list-style-type: none"><li>Built with Next.js, Firebase Firestore, and Google Auth.</li><li>GPA dashboards, real-time reviews, and auto-filled course data.</li></ul> | HackFax 2025   |
| <b>MindMate</b> – <i>AI Mental Wellness Companion</i> <ul style="list-style-type: none"><li>React + Azure OpenAI chatbot with journaling and mood prompts.</li></ul>                                                              | Feb – Mar 2024 |
| <b>Diabetic Retinopathy Detection</b> – <i>ML on Retinal Images</i> <ul style="list-style-type: none"><li>Used EfficientNet-B4, ResNet18, and custom CNN with QWK: 0.69.</li></ul>                                                | Apr – May 2023 |
| <b>Brain Hemorrhage Detection</b> – <i>Binary ResNet Classifier</i> <ul style="list-style-type: none"><li>Built ResNet18 to classify hemorrhagic vs normal CT scans.</li></ul>                                                    | Dec 2022       |

## Skills and Tools

**Languages:** Python, C, C++, SQL, Java, Bash  
**Libraries/Frameworks:** PyTorch, TensorFlow, Pandas, NumPy  
**Tools:** Docker, Kubernetes, GitHub Actions, Terraform  
**Web Dev:** HTML, CSS, JavaScript, React, Angular, jQuery  
**Cloud:** AWS, Azure, Google Cloud Platform  
**Concepts:** ML/DL, Security, UNIX Kernel, Data Structures

## Extracurricular Activities

- Vice President – Home Abroad (GMU).
- IEEE and Google Developer Student Club member.
- International Taekwondo Medalist.
- Student of the Year (3x).